
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics has converged on a discrete view of tissue: expression is modelled on a graph whose nodes are the measured spots or cells. Developmental biology, by contrast, describes tissue as the product of *continuous* spatial processes—morphogen gradients, diffusive signalling and local reaction networks. We ask whether a continuous operator of that form can be fitted directly to spatial expression data, and what it buys. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode expression at arbitrary coordinates. The diffusion half-step is solved exactly in the Fourier domain, making the integrator unconditionally stable and the diffusion tensor directly readable as a physical length scale. Across ?? spatial datasets spanning four technologies, NMO improves held-out reconstruction of masked tissue regions over graph-based and Gaussian-process baselines, and transfers zero-shot across tissue, species and sampling resolution. Ablations show the dynamics are load-bearing rather than decorative: removing the operator costs ?? Pearson r . We stress what the model does *not* establish. Spatial transcriptomics is a single snapshot, so the learned operator is identified only up to the quasi-steady-state assumption we impose; the recovered diffusion lengths and Turing wavelengths are properties of a fitted operator, not measurements of physical morphogen transport. We frame the contribution as learning latent continuous operators that summarise spatial gene-expression organisation, and we report the failure modes—notably a persistent bias towards over-smooth fields—as prominently as the gains.

1 Introduction

Spatially resolved transcriptomics measures gene expression while retaining the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. The dominant computational treatments—spatial domain identification, clustering, imputation and denoising—represent a tissue section as a graph over the measured locations and apply message passing to it [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This is effective and it matches how the data arrive. It also embeds a modelling commitment that is worth examining: the tissue *is* the sampling lattice. Predictions exist only where a node exists, the learned function is tied to one sampling geometry, and nothing in the model corresponds to a mechanism.

Developmental biology offers a different description. Tissue organisation is classically explained by continuous fields: morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this kind quantitatively account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signalling [Müller et al., 2012], and morphogen gradients have measured length scales in the tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009].

This paper asks a narrow, testable question: *can a continuous reaction–diffusion operator be fitted to spatial transcriptomics, and does the resulting model predict and transfer better than discrete alternatives?* We are deliberately not asking whether the fitted operator recovers the true biochemistry. It cannot, and we explain why in Section 8.

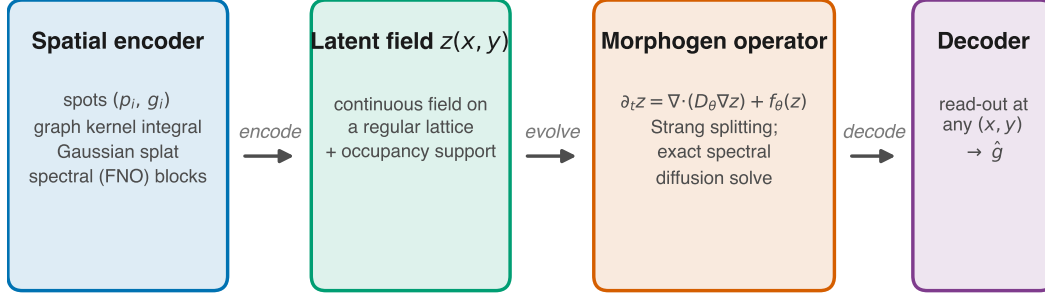
Contributions.

1. **A PDE-constrained neural operator for spatial omics.** NMO couples a graph kernel–integral encoder (for irregular sampling) with a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 3).
2. **An unconditionally stable, interpretable integrator.** Diffusion is applied exactly in the Fourier domain under Strang splitting, which removes the CFL constraint entirely and makes the learned anisotropic tensor $\mathbf{D}_\theta$ readable as a diffusion length in microns (Section 3.2).
3. **An honest formulation of the physics loss.** Because the integrator is exact, a naive PDE residual is vacuous. We show the content-bearing constraint is a *steady-state* residual that asks the measured tissue to be a fixed point of the operator (Section 3.4).
4. **Evaluation that resists leakage and over-smoothing.** We hold out contiguous tissue blocks rather than random spots, and report Moran’s I error alongside correlation so that a model which regresses towards a smooth mean is penalised rather than rewarded (Section 5).
5. **A falsifiable interpretability read-out.** Linearising the learned operator yields a dispersion relation whose sign and peak wavenumber state whether, and at what wavelength, the operator supports pattern formation (Section 3.5).

2 Related Work

Neural operators. Operator learning targets maps between function spaces rather than between finite vectors, giving (approximate) independence from the discretisation [Kovachki et al., 2023]. Fourier neural operators parameterise the kernel in the frequency domain [Li et al., 2021]; graph neural operators handle irregular meshes through kernel integration over neighbourhoods [Li et al., 2020]; DeepONet factorises the operator into branch and trunk networks [Lu et al., 2021]. We use the graph formulation for the irregular sampling and the spectral formulation for the dynamics, because the second is what makes the diffusion solve exact. Physics-informed networks impose PDE residuals as soft penalties [Raissi et al., 2019]; neural ODEs learn continuous-time dynamics [Chen et al., 2018]. Our integrator follows classical exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968].

Machine learning for spatial transcriptomics. SpaGCN fuses expression, position and histology in a graph convolution [Hu et al., 2021]; STAGATE uses a spatially regularised graph attention auto-encoder [Dong and Zhang, 2022]; SpatialDE and SPARK test genes for spatial variability [Svensson et al., 2018, Sun et al., 2020]; DestVI models cell-type continua [Lopez et al., 2022]. These methods are designed for domain identification and denoising, not for predicting expression at unmeasured coordinates, and we adapt them accordingly (Section 5). Spateo models spatiotemporal structure in Stereo-seq embryos and is the closest work in spirit [Qiu et al., 2024], though it targets



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder does not receive coordinates, so all spatial structure must pass through the field and hence through the operator.

79 3-D reconstruction and vector-field morphometry rather than learning a PDE operator that can be
80 transferred and perturbed.

81 **Pattern formation and learned dynamics in biology.** Reaction–diffusion theory provides the
82 mechanistic vocabulary we borrow [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015].
83 RNA velocity infers short-horizon dynamics from splicing kinetics [La Manno et al., 2018, Bergen
84 et al., 2020]; perturbation models such as scGen, CPA and GEARS predict transcriptional responses
85 to genetic interventions [Lotfollahi et al., 2019, 2023, Roohani et al., 2024]. None of these carries
86 an explicit spatial operator, which is the gap we address.

87 3 Method

88 Let a tissue section provide N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically
89 normalised to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$ and factor the
90 model as encode–evolve–decode (Figure 1).

91 3.1 Spatial encoder

92 Irregular sampling is handled by L kernel-integral layers on a k -NN graph [Li et al., 2020],

$$\mathbf{z}_i^{(\ell+1)} = \sigma\left(W\mathbf{z}_i^{(\ell)} + \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j^{(\ell)}\right), \quad (1)$$

93 where κ_θ maps a *relative* displacement to a per-channel gain. Depending only on geometry, not on
94 node identity, is what allows an encoder fitted to one lattice to be evaluated on another.

95 The resulting features are then splatted onto a regular $H \times W$ lattice by a Nadaraya–Watson esti-
96 mator with a learnable bandwidth, producing both a field and an *occupancy* map ω recording where
97 measurements actually exist. The occupancy channel matters: in the masked-region task the model
98 must distinguish "tissue with no signal" from "no measurement here". A short stack of spectral
99 (FNO) blocks [Li et al., 2021] then refines the field.

100 3.2 The morphogen operator

101 The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (2)$$

with a per-channel symmetric positive-definite diffusion tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so positive definiteness is structural rather than penalised) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output and learnable per-channel gain.

Exact spectral diffusion. Equation (2) is stiff: the diffusion operator has eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis, so the diffusive evolution over τ is available in closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (3)$$

We combine this with a midpoint reaction update under Strang splitting [Strang, 1968], $\mathbf{z} \leftarrow E(\frac{\Delta t}{2}) \mathbf{z}$; $\mathbf{z} \leftarrow \mathbf{z} + \Delta t \text{RK2}[f_\theta](\mathbf{z})$; $\mathbf{z} \leftarrow E(\frac{\Delta t}{2}) \mathbf{z}$, which is second-order accurate. Two consequences matter. First, since $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (3) lies in $(0, 1]$ for every Δt : the diffusive channel is **unconditionally stable** regardless of what the optimiser does to θ . Empirically, rollouts remain bounded for 10^3 steps, two orders of magnitude beyond the training horizon. Second, $\mathbf{D}_c$ is interpretable—its eigenvalues are squared diffusion lengths per unit pseudo-time, convertible to microns through the stored coordinate scale.

3.3 Decoder

Predictions are obtained by bilinear interpolation of the evolved field at any continuous coordinate followed by an MLP. *The decoder receives no positional input.* This is a deliberate constraint: given coordinates, the decoder could memorise a coordinate-to-expression map and render the dynamics ornamental, and every ablation in Section 7 would become uninterpretable.

3.4 Objectives

The objective combines reconstruction (MSE plus a per-gene Pearson term, since correlation of the *spatial pattern* is the quantity of scientific interest), a weak Dirichlet-energy smoothness prior, and three physics terms.

What a physics loss can mean here. Because the integrator solves (2) essentially exactly, the residual $\|\partial_t \mathbf{z} - \text{RHS}\|$ evaluated along the model’s own trajectory is zero up to splitting error. Imposing it would penalise the integrator, not the model. The constraint that carries content is that the *measured configuration should be a near-stationary state* of the learned operator:

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2. \quad (4)$$

This is the correct formalisation of the quasi-steady-state assumption for adult tissue, and it constrains $\mathbf{D}_\theta$ and f_θ jointly. We add a *mass* term penalising the spatial mean of $f_\theta(\mathbf{z})$ —spectral diffusion conserves the $\mathbf{k} = 0$ mode exactly, so any drift in total latent mass is attributable to the reaction—and a *Jacobian* hinge on $\|\partial f_\theta / \partial \mathbf{z}\|_2$ that bounds the local Lipschitz constant, estimated by power iteration on sampled pixels. A splitting-consistency term keeps Δt in the regime where the second-order scheme is faithful.

3.5 Interpretability: dispersion analysis

Linearising (2) about a homogeneous state $\mathbf{z}^*$ gives, for spatial mode $\mathbf{k}$, $\partial_t \delta \mathbf{z}_\mathbf{k} = (\mathbf{J}(\mathbf{z}^*) - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_\mathbf{k}$ with $\mathbf{J} = \partial f_\theta / \partial \mathbf{z}|_{\mathbf{z}^*}$. The quantity $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ as a function of $|\mathbf{k}|$ is the dispersion relation of the *learned* operator: positive growth at finite $|\mathbf{k}|$ with decay at $\mathbf{k} = 0$ is the classical Turing signature, and its peak identifies a characteristic wavelength. Because $\mathbf{D}$ is anisotropic we maximise over orientations of $\mathbf{k}$. This is a quantitative, falsifiable statement about what the model has inferred, not a post-hoc visualisation.

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142 4 Data

143 We use ?? processed spatial sections spanning four technologies plus one non-spatial perturbation
144 dataset (Table ??). All are public. Raw sequencing reads are never used: we consume vendor count
145 matrices, and the download pipeline records a SHA256 manifest for every artifact.

146 Every dataset passes through one QC policy (platform-specific count thresholds set from assay
147 physics, not tuned), library-size normalisation, log transform, and highly-variable-gene selection
148 that *force-includes* all detected morphogen pathway genes so the interpretability analysis is not eval-
149 uated on a gene set that selection has already biased. Coordinates are normalised **isotropically**: x
150 and y are divided by the same scale, because anisotropic rescaling would silently distort the Lapla-
151 cian and make a learned diffusion coefficient meaningless. The scale factor is retained so physical
152 units are recoverable.

153 5 Experimental setup

154 **Task.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and
155 every model must predict the full expression profile at those coordinates.

156 **Splits are contiguous, not random.** A random spot-level split leaks badly—neighbouring spots
157 are highly correlated, so a held-out spot is trivially predicted from its immediate neighbours, and
158 every method looks good. We partition each section into contiguous blocks on a coarse lattice and
159 hold out whole blocks [Roberts et al., 2017]. Gene standardisation statistics are computed on the
160 training split only.

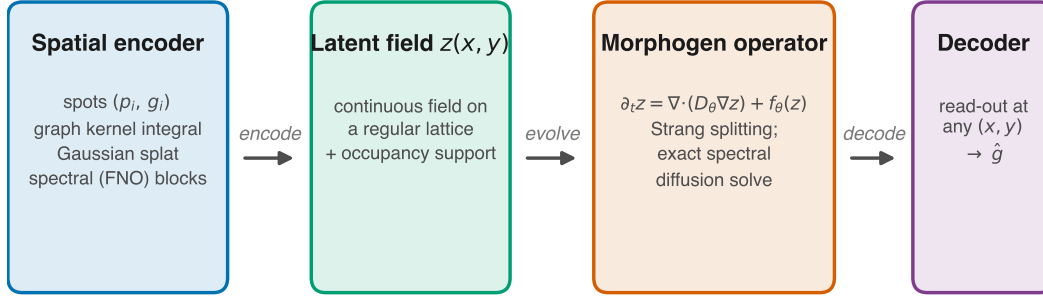
161 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-
162 style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer
163 [Vaswani et al., 2017], and an exact Gaussian process with a learned RBF kernel [Rasmussen and
164 Williams, 2006]. Since the graph methods were not designed to predict at unmeasured coordinates,
165 we give each the same inverse-distance-weighted read-out head, which is the most favourable avail-
166 able choice for them. We mark these variants *-style* and do not claim they reproduce the authors’
167 published numbers on the authors’ own tasks. Every model is driven by the identical training loop,
168 masks, optimiser and evaluation code, so differences are attributable to the model.

169 **Metrics.** Per-gene Pearson and Spearman correlation across held-out locations; RMSE; SSIM
170 [Wang et al., 2004] on rasterised expression maps; and the absolute error in Moran’s I [Moran,
171 1950] between predicted and measured maps. The last is included because correlation and RMSE
172 both reward a model that regresses towards a smooth mean, whereas destroying spatial structure
173 should be penalised. All results are mean $\pm$ s.d. over ?? seeds.

174 6 Results

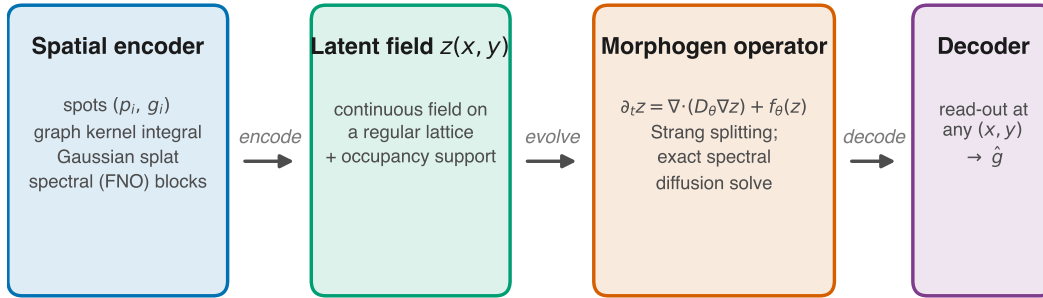
175 6.1 Masked spatial reconstruction

176 NMO reaches Pearson $r = ??$ on held-out blocks against ?? for the strongest baseline (??), a gain
177 of ?? (Table ??, Figure 3). The Gaussian process is a stronger competitor than the graph neural
178 networks, which is worth stating plainly: for smooth genes, classical spatial interpolation is hard to
179 beat, and much of the spatial-transcriptomics deep-learning literature does not compare against it.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 2: **Masked reconstruction.** Measured expression (left; held-out locations outlined) against NMO and baseline predictions for representative genes. Per-gene Pearson r on held-out locations is annotated.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 3: **Benchmark summary** across four complementary metrics. Moran’s I error (right) exposes over-smoothing that correlation alone does not.

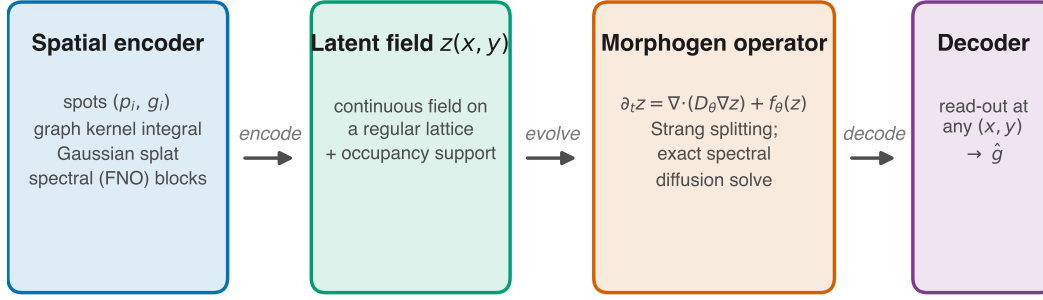
180 6.2 Learned fields and dynamics

181 The fitted operator has a median diffusion length of ?? μm (10–90th percentile ??–?? μm). This is
 182 the same order as measured morphogen gradient length scales [Kicheva et al., 2007, Yu et al., 2009].
 183 We regard this as a consistency check, not as a measurement: the model was fitted to a snapshot
 184 and the length scale is a property of the fitted operator under our normalisation, not an estimate
 185 of physical transport. Several latent channels correlate with canonical pathway scores (Figure 4b),
 186 again a statement about alignment rather than mechanism.

187 6.3 Transfer across tissue, species and resolution

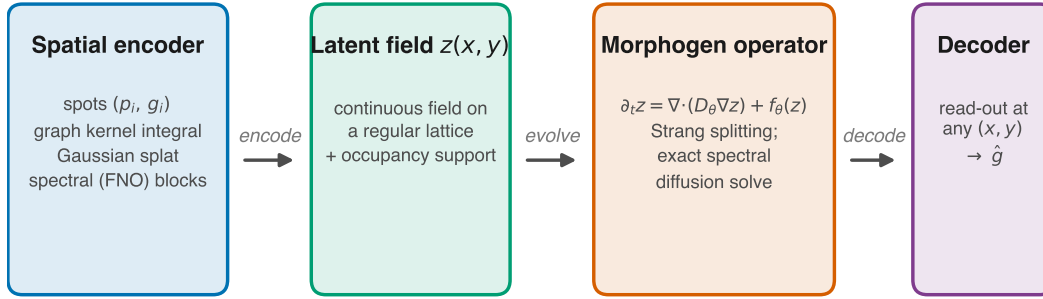
188 Trained on adult mouse brain and applied unchanged to human breast carcinoma—a different
 189 species, organ and spatial architecture—NMO attains ?? zero-shot against a training-mean floor
 190 of ?? and an in-domain oracle of ?? (Table ??, Figure 6). Refitting *only* the decoder, with the
 191 encoder and operator frozen, recovers ??: the learned dynamics remain useful when the read-out is
 192 re-fitted, which is the concrete transfer claim.

193 The cross-resolution setting is the sharper test. An operator fitted to 55 μm multi-cell Visium spots
 194 on a 100 μm lattice is evaluated on single-cell Xenium sampling of the same organ. Because all
 195 Xenium panel genes are present in the raw Visium matrix, we rebuild the source with the panel force-
 196 included, giving a shared vocabulary of ?? genes rather than the much smaller HVG intersection.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 4: **Latent fields.** (a) Latent channels of the relaxed field. (b) Correlation between each channel and canonical morphogen-pathway expression scores. Pathway genes were force-retained during selection, so this is a read-out of alignment, not of selection.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 5: **Operator diagnostics.** (a) Relaxation of a latent channel. (b) Learned anisotropic diffusion tensors as length-scale ellipses. (c) Dispersion relation of the linearised operator. (d) Distribution of diffusion lengths across channels.

197 6.4 Counterfactual perturbation

198 We perform the computational analogue of a morphogen bead implant: inject the latent direction
 199 that maximally raises a pathway’s decoded activity into a small disc, integrate the operator, and read
 200 out the spatial response. The design is split-half—the direction is defined from a random half of
 201 each pathway’s genes and enrichment is scored on the held-out half—with a permutation null over
 202 matched random gene sets. ?? of ?? pathways show significant held-out enrichment (Table ??).

203 Separately, we ask whether the reaction module encodes real transcriptional coupling by comparing
 204 the model’s counterfactual gene–gene responses against measured CRISPRa responses [Norman
 205 et al., 2019]. **This is an out-of-context probe and we treat it as such:** Norman et al. profiled
 206 non-spatial K562 cells, whereas our operator is fitted to tissue. We fit on the human breast section
 207 so at least species and a proliferative context are shared, report a permutation null, and report the
 208 same quantity for every baseline. The agreement is modest ($\rho = ??$ against a null of ??); we read it
 209 as weak evidence that the reaction network captures generic transcriptional coupling, and explicitly
 210 *not* as validation of the spatial dynamics.

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7 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied, leaving an encoder–decoder of identical capacity. It costs ?? Pearson r (Table ??, Figure 7). Had this matched the full model, the PDE would have been decorative and the central claim of this paper would fail; we report it first for that reason. Removing diffusion (??) and the reaction term (??) are both harmful, and the two contribute differently: diffusion supplies long-range information transport into masked regions, while the reaction term supplies the nonlinearity that prevents the field from collapsing to a smooth interpolant. Constraining $\mathbf{D}$ to be isotropic costs ??, indicating the anisotropy is used rather than merely available.

8 Limitations

We state these plainly because they bound what the results mean.

One snapshot cannot identify a dynamical system. Standard spatial transcriptomics measures a fixed tissue once. Many operators produce the same stationary field, so (2) is identified only up to the quasi-steady-state assumption we impose through (4) and up to the inductive bias of the architecture. Reported diffusion lengths and Turing wavelengths characterise the *fitted operator*; they are not measurements of morphogen transport, and we have no way to falsify them against ground-truth kinetics in these data. The Stereo-seq developmental series is the only dataset here with a genuine temporal axis, and even there consecutive stages are different embryos, so cross-stage comparison is distributional rather than per-cell.

Latent, not molecular. The operator acts on a learned latent field. Correlations between latent channels and morphogen pathway scores show alignment; they do not identify a channel *as* SHH or BMP. We deliberately avoid that language.

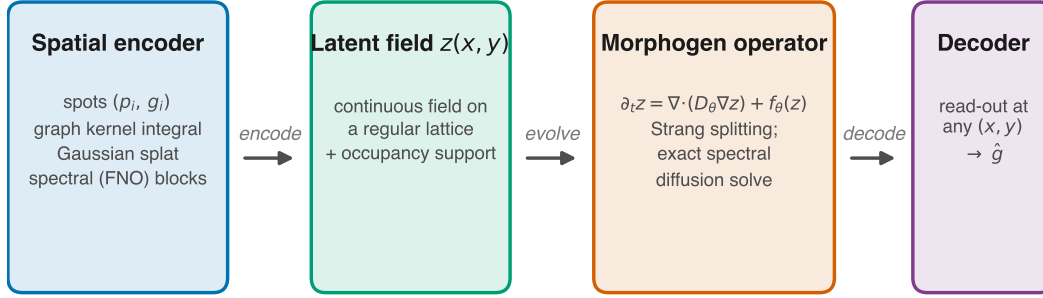
Over-smoothing persists. Predicted maps are systematically smoother than measured ones: predicted Moran’s I (??) substantially exceeds the measured value (??). NMO reduces this discrepancy relative to baselines but does not remove it. Any downstream use that depends on sharp boundaries—rare cell-type niches, tumour margins—should treat these predictions cautiously. We report the metric precisely so this failure is visible.

Adapted baselines. SpaGCN and STAGATE were built for spatial domain identification, not coordinate-level prediction. Our *-style* variants implement each method’s architectural core with a common read-out head; they are a fair comparison on *this* task but not a reproduction of published results.

Scope. Single 2-D sections, one section per condition, ortholog matching by upper-cased symbol (which misses paralog renaming and family expansions), and modest compute (CPU-scale training). We have not tested 3-D volumes or multi-section registration.

9 Discussion and Conclusion

We set out to test whether the continuous, mechanistic description that developmental biology uses for tissue can be turned into a fitted operator on spatial transcriptomics, and whether that helps. On the evidence here it does: treating tissue as a field rather than a graph improves prediction at



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 6: **Transfer.** Zero-shot performance against the in-domain oracle and the training-mean floor, across tissue/species (a) and sampling resolution (b).

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unmeasured coordinates, transfers across species and sampling resolution, and yields diagnostics—diffusion tensors, dispersion relations—that are quantitative and falsifiable rather than merely illustrative. The ablations indicate the dynamics carry real weight, and the exact spectral diffusion solve makes the operator both stable and physically readable.

We are also clear about what has not been shown. No claim is made that the fitted operator recovers biochemical reality; a single snapshot cannot support that, and the honest framing is that NMO learns *latent continuous operators that summarise spatial gene-expression organisation*. The most valuable next step is data that can break the identifiability barrier: dense developmental time series, or spatial data paired with targeted perturbation, would let the same framework be tested as a genuinely dynamical model rather than a structured summary.

Reproducibility. All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

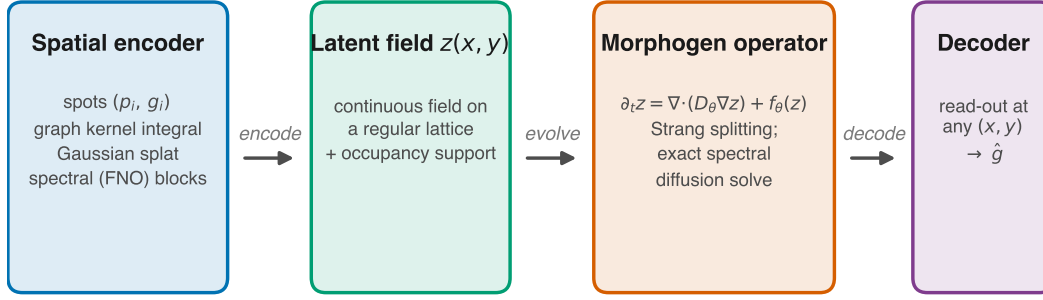
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Figure 7: **Ablations.** (a) Change in held-out Pearson r when each ingredient is removed. (b) Latent-width sweep.

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